

Smart Parking Gate System

Project Aim

Design an automated smart parking gate for secure vehicle access..

Project Overview

The Smart Parking Gate System is an Arduino-based automation project that controls vehicle entry using sensors and a servo motor. It detects vehicles, confirms motion, and identifies day or night conditions. The system automatically operates the gate, provides audio alerts, displays status messages on an LCD, and ensures efficient parking management.

Components Used

- Arduino Uno
- HC-SR04 Ultrasonic Sensor
- PIR Motion Sensor
- LDR
- Servo Motor (SG90)
- Piezo Buzzer
- 16×2 LCD
- 10kΩ Potentiometer
- 2 Status LEDs

- Green (Gate Open)
- Red (Gate Closed)

- 5 Component Health LEDs (Green)
- Ultrasonic OK
- LDR OK
- PIR OK
- Servo OK
- Buzzer OK
- Resistors (220Ω)

Construction and Wiring Details

Component	Arduino Pin(s)	Additional Connections	Purpose
HC-SR04 Ultrasonic Sensor	TRIG → D2, ECHO → D3	VCC → 5V, GND → GND	Detects vehicles approaching the gate
PIR Motion Sensor	OUT → D4	VCC → 5V, GND → GND	Confirms motion near the gate
LDR (Photoresistor)	A0	Terminal 1 → 5V, Terminal 2 → A0, 10kΩ resistor from A0 → GND	Determines day/night conditions
Servo Motor	Signal → D5	VCC → 5V, GND → GND	Opens and closes the parking gate
Piezo Buzzer	D6	Negative terminal → GND	Provides alerts and warning sounds
Gate Open LED (Green)	D7	220Ω resistor in series, Cathode → GND	Indicates gate is open
Gate Closed LED (Red)	D8	220Ω resistor in series, Cathode → GND	Indicates gate is closed
Ultrasonic Status LED	D9	220Ω resistor in series, Cathode → GND	Indicates ultrasonic sensor is functioning
LDR Status LED	D10	220Ω resistor in series, Cathode → GND	Indicates LDR is functioning
PIR Status LED	D11	220Ω resistor in series, Cathode → GND	Indicates PIR sensor is functioning
Servo Status LED	D12	220Ω resistor in series, Cathode → GND	Indicates servo motor is functioning
Buzzer Status LED	D13	220Ω resistor in series, Cathode → GND	Indicates buzzer is functioning
LCD Display (16×2)	RS → A1, E → A2, DB4 → A3, DB5 → A4, DB6 → A5, DB7 → D1	RW → GND, VCC → 5V, GND → GND	Displays greetings and gate status messages

Component	Arduino Pin(s)	Additional Connections	Purpose
Potentiometer (10kΩ)	V0 (LCD Contrast)	One end → 5V, Other end → GND, Middle pin → LCD V0	Adjusts LCD display contrast

Code Structure / Program Flow

Start



Read LDR



Determine Day/Night



Read PIR



Read Ultrasonic

Vehicle <20 cm ?



NO



Continue Monitoring

YES



PIR Detected?



NO



Continue Monitoring

YES

|

Day?

|

YES -----→ Open Gate

|

NO

|

Sound Buzzer 2 sec

|

Open Gate

Gate Open 3 sec

Close Gate

Vehicle Still Present >5 sec?

YES

|

Continuous Alarm

|

Wait until clear

NO

|

Continue Monitoring.

Source Code

```
#include <Servo.h>
```

```
#include <LiquidCrystal.h>
```

```
Servo gate;
```

```
// LCD Pins
```

```
LiquidCrystal lcd(A1, A2, A3, A4, A5, 1);
```

```
// Ultrasonic
```

```
const int trigPin = 2;
```

```
const int echoPin = 3;
```

```
// PIR
```

```
const int pirPin = 4;
```

```
// Servo
```

```
const int servoPin = 5;
```

```
// Buzzer
```

```
const int buzzer = 6;
```

```
// Gate LEDs
```

```
const int greenGate = 7;
```

```
const int redGate = 8;
```

```
// Health LEDs
```

```
const int ultraLED = 9;
```

```
const int ldrLED = 10;  
const int pirLED = 11;  
const int servoLED = 12;  
const int buzzerLED = 13;
```

```
// LDR
```

```
const int ldrPin = A0;
```

```
long duration;
```

```
float distance;
```

```
unsigned long vehicleStartTime = 0;
```

```
void setup()
```

```
{
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  pinMode(pirPin, INPUT);
```

```
  pinMode(buzzer, OUTPUT);
```

```
  pinMode(greenGate, OUTPUT);
```

```
  pinMode(redGate, OUTPUT);
```

```
  pinMode(ultraLED, OUTPUT);
```

```
pinMode(ldrLED, OUTPUT);  
pinMode(pirLED, OUTPUT);  
pinMode(servoLED, OUTPUT);  
pinMode(buzzerLED, OUTPUT);
```

```
gate.attach(servoPin);  
gate.write(0);
```

```
lcd.begin(16, 2);
```

```
digitalWrite(redGate, HIGH);
```

```
lcd.print("SMART PARKING");  
lcd.setCursor(0, 1);  
lcd.print("SYSTEM READY");
```

```
delay(3000);  
lcd.clear();  
}
```

```
void loop()  
{  
  // Health LEDs  
  digitalWrite(ultraLED, HIGH);  
  digitalWrite(ldrLED, HIGH);  
  digitalWrite(pirLED, HIGH);
```

```
digitalWrite(servoLED, HIGH);  
digitalWrite(buzzerLED, HIGH);  
  
int ldrValue = analogRead(ldrPin);  
bool night = (ldrValue < 500);  
  
// Greeting  
if (night)  
{  
    lcd.setCursor(0, 0);  
    lcd.print("GOOD EVENING ");  
}  
else  
{  
    lcd.setCursor(0, 0);  
    lcd.print("GOOD MORNING ");  
}  
  
// Ultrasonic  
digitalWrite(trigPin, LOW);  
delayMicroseconds(2);  
  
digitalWrite(trigPin, HIGH);  
delayMicroseconds(10);  
  
digitalWrite(trigPin, LOW);
```



```
duration = pulseIn(echoPin, HIGH);
```

```
distance = duration * 0.034 / 2;
```

```
bool motion = digitalRead(pirPin);
```

```
if (distance <= 20 && motion)
```

```
{
```

```
  lcd.setCursor(0, 1);
```

```
  lcd.print("VEHICLE FOUND ");
```

```
  if (vehicleStartTime == 0)
```

```
    vehicleStartTime = millis();
```

```
  // Night warning
```

```
  if (night)
```

```
  {
```

```
    lcd.clear();
```

```
    lcd.print("NIGHT MODE");
```

```
    tone(buzzer, 1000);
```

```
    delay(2000);
```

```
    noTone(buzzer);
```

```
  }
```

```
openGate();

// Vehicle blocking gate
while (distance <= 20)
{
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;

    if (millis() - vehicleStartTime > 5000)
    {
        lcd.clear();
        lcd.print("REMOVE VEHICLE");
        lcd.setCursor(0, 1);
        lcd.print("GATE BLOCKED");

        tone(buzzer, 800);
    }
}
```

```
    if (distance > 20)
    {
        noTone(buzzer);
        vehicleStartTime = 0;
        break;
    }
}
```

```
    closeGate();
}
else
{
    vehicleStartTime = 0;

    lcd.setCursor(0, 1);
    lcd.print("GATE READY  ");
}
```

```
    delay(200);
}
```

```
void openGate()
{
    digitalWrite(redGate, LOW);
    digitalWrite(greenGate, HIGH);
}
```

```
lcd.clear();  
lcd.print("GATE OPEN");  
lcd.setCursor(0, 1);  
lcd.print("WELCOME");
```

```
gate.write(90);
```

```
delay(3000);  
}
```

```
void closeGate()  
{  
  lcd.clear();  
  lcd.print("GATE CLOSING");  
  lcd.setCursor(0, 1);  
  lcd.print("PLEASE WAIT");
```

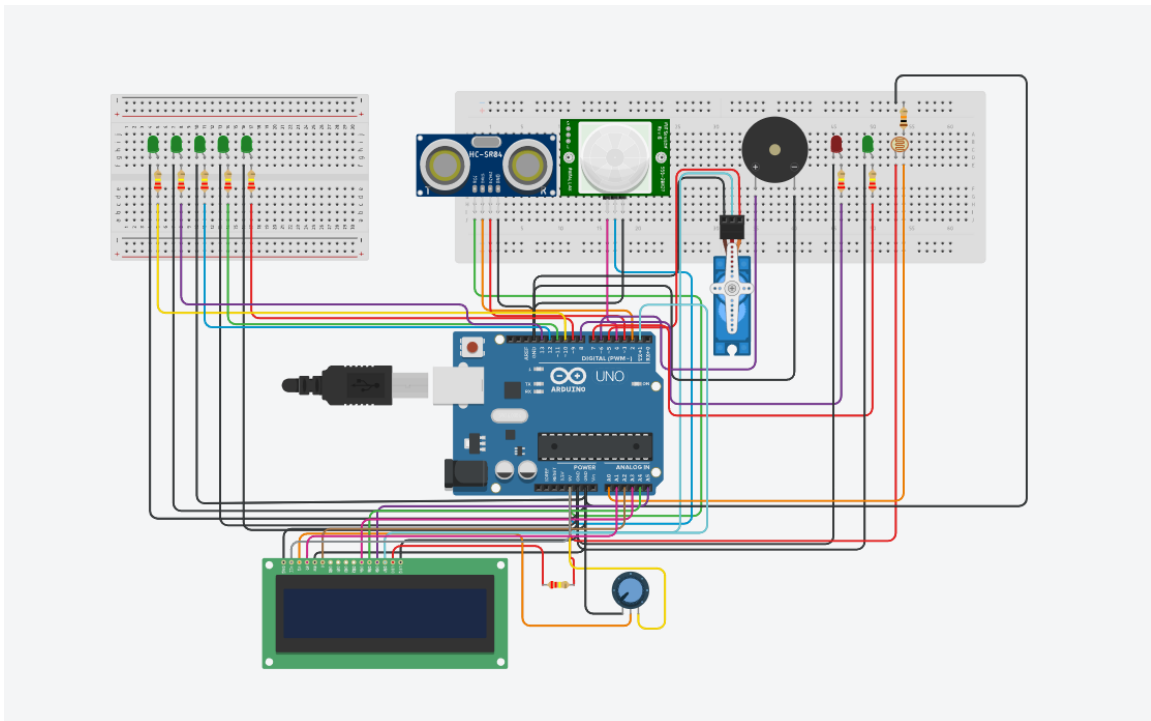
```
delay(1000);
```

```
gate.write(0);
```

```
digitalWrite(greenGate, LOW);  
digitalWrite(redGate, HIGH);
```

```
lcd.clear();  
}.
```

Project Photograph



Links

Google drive :

https://drive.google.com/drive/folders/1GEi2cqGkShexGLSm9h5sOYaS0CmqDtWS?usp=drive_link

TinkerCad :

https://www.tinkercad.com/things/6J7J2iu2PiZ/editel?returnTo=%2Fdashboard&sharecode=DAf60AfZr-BKVl7YNN0qVloo_Jb3vxt9bV7eLmKw5Ho

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>